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IDENTIFICATION OF HEALTH RISK AND IMPACT ASSESSMENT

Abstract

Various examples of techniques for assessing patient risk and patient impact for health interventions are disclosed. In one example, a computer implemented method is disclosed that includes obtaining from a patient database patient information for a plurality of patients, the patient information including at least a risk score, clustering by a processor the plurality of patients based on the patient information using a categorization model, where the categorization model is trained using unsupervised learning, and assigning by the processor a risk category to each patient based on the clustering.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority pursuant to 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/560,309, filed Mar. 1, 2024, entitled “Identification of Health Risk and Impact Assessment,” which is hereby incorporated by reference herein in its entirety. This application is related to U.S. application Ser. No. _____ [[Attorney Docket No. P316213.US.02]], filed on Feb. 28, 2025, entitled “Transition of Care Work Flow and Prioritization System,” which is hereby incorporated by reference herein in its entirety for all purposes.

FIELD

[0002] The present disclosure relates generally to systems for categorizing patients based on health risks.

BACKGROUND

[0003] Community health programs and patient outreach can significantly improve patient health outcomes. Patients with higher health needs benefit from more intensive care management through such programs. Identifying patients with higher health needs can be challenging. Various methods of identifying such patients are often subject to biases, which may result in under identification of some higher health needs patients, along with poorer health outcomes for such patients.

SUMMARY

[0004] In one example, a computer implemented method is disclosed. The method includes obtaining from a patient database patient information for a plurality of patients, the patient information including at least a risk score, clustering by a processor the plurality of patients based on the patient information using a categorization model, where the categorization model is trained using unsupervised learning, and assigning by the processor a risk category to each patient based on the clustering.

[0005] In another embodiment, a method is disclosed that includes obtaining patient information for a plurality of patients, the patient information including at least a risk score and a diagnosis for each patient, clustering the plurality of patients based on the patient information using a categorization model, where the categorization model is trained using unsupervised learning, assigning a risk category to each patient based on the clustering, identifying of reach patient an impact of at least one intervention based on the diagnosis and using an impact model, and prioritizing at least a patient of the plurality of patients based on the risk category and the impact.

[0006] In yet another embodiment, a non-transitory computer readable media encoded with instructions, which when executed by one or more processors of a segmentation system is disclosed. The instructions cause the segmentation system to obtain patient information for a plurality of patients, the patient information including at least a risk score, cluster the plurality of patients based on the patient information using a categorization model, where the model is trained using unsupervised learning, and assigning a risk category to each patient based on the clustering.

[0007] In another embodiment, a computer implemented method is disclosed. The method includes obtaining patient information for a plurality of patients, the patient information including at least a diagnosis, determining using an impact model and based on the diagnosis an impact of at least one intervention based on the diagnosis, and prioritizing at least a patient of the plurality of patients based on the impact of the at least one intervention.

[0008] In another embodiment, a computer implemented method is disclosed. The method includes receiving a request for a patient profile, determining a risk category for the patient based at least on

a risk score, where the risk category is determined using an unsupervised learning model, and presenting via a user interface the patient profile where the patient profile includes a risk category for the patient.

[0009] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. A more extensive presentation of features, details, utilities, and advantages of the present invention as defined in the claims is provided in the following written description of various embodiments and implementations and illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 illustrates an example segmentation system and various systems in communication with the segmentation system through a network, in accordance with various embodiments of the disclosure.

[0011] FIG. 2 illustrates an example segmentation system in accordance with various embodiments of the disclosure.

[0012] FIG. 3 illustrates an example user interface of the segmentation system.

[0013] FIG. 4 illustrates an additional example user interface of the segmentation system.

[0014] FIG. 5 is a schematic diagram of an example computer system implementing various embodiments in the examples described herein.

[0015] FIG. 6 illustrates an example process for assigning risk categories to patients using the segmentation system.

[0016] FIG. 7 illustrates an example process for assessing impact of an intervention for a patient using the segmentation system.

[0017] FIG. 8 illustrates an example process for assigning risk category to a patient and assessing impact of an intervention for the patient using the segmentation system.

DETAILED DESCRIPTION

[0018] Preventative healthcare, such as targeted outreach efforts, specialized programs, and the like generally decrease healthcare costs while improving health outcomes of patients. For example, patients participating in a program targeting management of a chronic condition may have fewer health complications related to the condition and may accrue less costs to treat the condition. However, identifying patients who are likely to most benefit from such efforts can be difficult. For example, identification of patients based on diagnosis codes (e.g., as utilized by insurance companies) may result in the inclusion of patients with well controlled chronic health conditions as high risk when such patients are less likely to benefit from outreach efforts. Identification of patients based on past healthcare utilization (e.g., by healthcare costs incurred) may result in the exclusion of patients with uncontrolled or poorly controlled chronic health conditions who have not historically utilized healthcare services, leading to poorer outcomes for those patients who may be excluded from targeted outreach efforts.

[0019] Further, existing models and systems of patient risk assessment may be inherently biased (e.g., biased based on race, gender, or other factors). Machine learning (ML) models may generally be less biased than manual or human implemented risk assessment. However, ML models may retain biases, which may decrease accuracy of the models. Some bias mitigation methods may identify underrepresented groups and may adjust outputs for such groups in some manner intended to mitigate potential bias. However, such methods are ineffective where data regarding membership in an underrepresented group is unavailable. Further, some types of data, such as healthcare

utilization and cost of care, may be inherently biased against certain groups of individuals due to structural or societal inequities. Accordingly, models trained with and utilizing such data to categorize patients are inherently biased. Biased models are generally less accurate, such that the patients who would most benefit from targeted efforts may not be properly identified, resulting in poorer health outcomes and higher cost of care when compared to unbiased models.

[0020] The systems and methods described herein improve identification of patients who have a high likelihood of utilization of healthcare services and/or who may benefit from outreach efforts (e.g., by improving health outcomes, lowering healthcare costs, and the like). Models used herein may utilize multiple types of risk scores as input, generally reducing bias which would likely be present with manual sorting or utilizing models with different types of input. Healthcare utilization metrics used herein may also include inpatient events identified by both claims and admission-discharge-transfer (ADT) data (e.g., including data related to one or more admission, discharge, and/or transfer events) including overlapping events (e.g. one claim/health information exchange (HIE) event starts before another claim/HIE event has ended) for more accuracy. Elimination of bias generally more accurately categorizes patients, reducing overall healthcare costs for groups of patients and improving health outcomes through targeted outreach, preventative care, and/or other interventions.

[0021] Categorization of patients as described herein may result in cost savings and improve health outcomes for community health organizations and other organizations managing medical care for large groups of patients. For example, such organizations may tailor outreach and management based on the risk level of patients. The correct categorization of low risk patients may result in cost savings, as those patients generally do not need the higher, more costly levels of care. The correct categorization of high risk patients may result in improved health outcomes, as those patients are provided with more intensive care management and outreach.

[0022] The systems and methods described herein further identify patients who are particularly likely to benefit from inclusion in targeted programs or from increased outreach. For example, specific programs or care protocols may be used to manage behavioral health conditions which are not well controlled. The methods described herein may identify patients diagnosed with behavioral conditions who are most likely to see improvement from inclusion in such programs. Such identification may both improve medical outcomes for the patient population as a whole (e.g., by including patients who are likely to benefit from such programs) and control costs associated with such programs (e.g., by excluding patients who are less likely to benefit from such programs). The targeted outreach, follow-up, and/or other services identified may be assigned (e.g., dispatched) based on the results of the segmentation so as to treat those patients that are the highest risk and may be the most likely to benefit from the services (or other categories/weightings used by the segmentation system). It should be noted that various embodiments may be used to provide a general risk calculation that may be displayed on a user interface, e.g., top ten patient identifiers may be displayed to a nurse or may be stored within a patient database used for automated services follow-up.

[0023] Various embodiments of the present disclosure will be explained below in detail with reference to the accompanying drawings. Other embodiments may be utilized, and structural, logical and electrical changes may be made without departing from the scope of the present disclosure.

[0024] Turning now to the figures, FIG. 1 illustrates an example segmentation system **102** and various systems in communication with the segmentation system **102** through a network **110**. As shown in FIG. 1, the segmentation system **102** may be accessed through a network **110** by various user devices **104**. The segmentation system **102** may also communicate with various care systems **106** and data storage **108** through the network **110**. For example, the segmentation system **102** may access various forms of patient data, information, risk scores, and the like from various care systems **106** and/or data storage **108** locations. The segmentation system may also communication

with these same systems **106** and/or data storage locations **108**, such as to output segmentation information for various patient populations to health services databases, provider databases, or the like.

[0025] The segmentation system **102** may generally utilize patient information to identify subgroups of patients or individual patients most likely to benefit from targeted intervention, outreach, and other preventative healthcare interventions. Such identification may include grouping patients based on health risks, such as likelihood of utilization of healthcare. In such examples, membership in a higher risk group may indicate that an individual or patient may benefit from additional resources and outreach. Further, the segmentation system **102** may assess which patients are most likely to be positively impacted by various outreach efforts or interventions.

[0026] In some examples, output from the segmentation system **102** may be utilized to prioritize patients or individuals for outreach or other healthcare interventions. For example, patients with a high risk of utilization and a high likelihood of positive impact of healthcare interventions may be contacted first for such interventions. In some examples, risk groups may be utilized to identify a course of action for an individual patient. For example, a patient in a low risk group may be contacted quarterly to discuss healthcare goals and may be scheduled for a yearly appointment with a primary care provider. A patient in a high risk group may be contacted more frequently, may be scheduled with various providers including specialists, and may be enrolled in targeted programs related to particular healthcare conditions. The cadence of the contact may be determined and output by the system to a user who can complete the actual outreach or may be automated, e.g., texts, calls, or the like.

[0027] The segmentation system **102** may be generally implemented by a computing device or combinations of computing resources in various embodiments. In various examples, the segmentation system **102** may be implemented by one or more servers, cloud computing resources, and/or other computing devices. The segmentation system **102** may, for example, utilize various processing resources to categorize and/or prioritize patients based on various patient data, e.g., updating a ranking of services for a patient database based on the segmentation system, flagging a patient in a patient database or services database for different or earlier services, or the like. The segmentation system **102** may further include memory and/or storage locations to store program instructions for execution by the processor and various data utilized by the segmentation system **102**. The segmentation system **102** may be used to generate “call” or follow-up lists of patients based on provider or health services grouping to allow certain patients to be contacted by health providers earlier than others. The output lists may be stored in a database, transmitted via one or more notifications to various devices, and/or provided via a user interface.

[0028] User devices **104** may be utilized to provide input to, and receive output from, the segmentation system **102**. For example, a user device **104** may belong to an end user tasked with patient outreach efforts, such as a health services provider. The user device **104** may be utilized to request categorization of a particular patient (e.g., placement of the patient into a particular segment), prioritization or categorization of a group of patients, and the like. The user device **104** may receive categorizations, identifications of particular programs or outreach efforts for a patient, and other output from the segmentation system **102**.

[0029] Generally, the user device **104** and/or other user devices in communication with the segmentation system **102** may be devices belonging to an end user utilizing the segmentation system **102**, such as end users tasked with patient outreach efforts and the like. In some implementations, many user devices **104** may access the segmentation system **102** to view a segment or grouping for a patient, prioritize patients in a group of patients, view interventions likely to have a positive impact on a patient, and perform other functions. In various embodiments, user devices **104** may be authenticated by an authentication service prior to accessing the segmentation system **102**.

[0030] In various implementations, the user device **104** and/or other user devices in communication

with the segmentation system **102** may be implemented using any number of computing devices including, but not limited to, a computer, a laptop, mobile phone, smart phone, wearable device (e.g., AR/VR headset, smart watch, smart glasses, or the like), smart speaker, vehicle (e.g., automobile), or appliance. Generally, the user devices may include one or more processors, such as a central processing unit (CPU) and/or graphics processing unit (GPU). The user devices may generally perform operations by executing executable instructions (e.g., software) using the processor(s).

[0031] A care system **106** may be in communication with the segmentation system **102**. In various examples, multiple care systems **106** may be in communication with the segmentation system **102** via the network **110**. Various care systems **106** in communication with the segmentation system **102** may be associated with or utilized by insurance companies, healthcare providers, healthcare facilities, or other healthcare related entities. For example, care systems **106** may include electronic health record (EHR) systems, insurance claim systems, and other systems storing or utilizing patient data. In some examples, care systems **106** may further be systems associated with entities providing interventions or administering programs recommended by the segmentation system **102**. Such care systems **106** may utilize output data from the segmentation system **102**.

[0032] Various data utilized by the segmentation system **102** may be stored at data storage **108**. For example, various types of patient information or patient data may be stored at data storage **108**. Patient profiles, including patient age, diagnoses, categorizations, and/or additional patient data may be stored at data storage **108**. The segmentation system **102** may both retrieve data from and write data to data storage **108**.

[0033] The network **110** may be implemented using one or more of various systems and protocols for communications between computing devices. In various embodiments, the network **110** or various portion of the network **110** may be implemented using the Internet, a local area network (LAN), a wide area network (WAN), and/or other networks. In addition to traditional data networking protocols, in some embodiments, data may be communicated according to protocols and/or standards including near field communication (NFC), Bluetooth, cellular connections, and the like.

[0034] Components of the segmentation system **102** and those in communication with the segmentation system **102** shown in FIG. 1 are exemplary and may vary in some embodiments. For example, in some embodiments, the segmentation system **102** may be distributed across multiple computing elements, such that components of the segmentation system **102** communicate with one another through the network **110**. Further, in some embodiments, computing resources dedicated to the segmentation system **102** may vary over time based on various factors such as usage of the segmentation system **102**. In some embodiments, the segmentation system **102** may communicate with multiple user devices **104** and/or other systems not shown in FIG. 1.

[0035] FIG. 2 illustrates an example segmentation system **102**. The segmentation system **102** generally stratifies, triages, or sorts patients according to predicted intensity of medical care and types of needs. Such stratification may help to prioritize preventative care and outreach to patients with higher medical or behavioral health needs. For example, the segmentation system **102** may categorize a patient or group of patients based on likelihood of healthcare utilization (e.g., low likelihood, rising likelihood, and high likelihood) based on risk scores and other patient information.

[0036] In various examples, the segmentation system **102** may include or utilize one or more hosts or combinations of compute resources, which may be located, for example, at one or more servers, cloud computing platforms, computing clusters, and the like. Generally, the segmentation system **102** is implemented by compute resources including hardware for memory **118** and one or more processors **116**. For example, the segmentation system **102** may utilize or include one or more processors, such as a CPU, GPU, and/or programmable or configurable logic.

[0037] In some embodiments, various components of the segmentation system **102** may be

distributed across various computing resources, such that components of the segmentation system **102** communicate with one another through the network **110** and/or using other communications protocols. For example, in some embodiments, the segmentation system **102** may be implemented as a serverless service, where computing resources for various components of the segmentation system **102** may be located across various computing environments (e.g., cloud platforms) and may be reallocated dynamically and/or automatically according to, for example, resource usage of the segmentation system **102**. In various implementations, the segmentation system **102** may be implemented using organizational processing constructs such as functions implemented by worker elements allocated with compute resources, containers, virtual machines, and the like.

[0038] In various examples, the memory **118** may directly store various types of data and/or instructions for accessing such data from another location or source. For example, memory **118** may store patient data **130**. Patient data **130** may generally be used to categorize patients, identify patients for inclusion in particular programs, generate user interfaces to the segmentation system **102**, and the like. In various examples, patient data may include biographical data (e.g., name, age, contact information, sex, and the like), treatment data (e.g., information about healthcare appointments, diagnoses, providers, and the like), intervention data (e.g., contacts with the patients, interventions already given to the patient, and the like), insurance data, risk scores, assigned segment, and the like. Such patient data **130** may be utilized by the segmentation system **102** to categorize or assign patients to segments, prioritize patients for particular interventions, present profiles or user interfaces to the segmentation system **102**, and to perform other functions of the segmentation system **102**. In addition to be displayed on various user interfaces, such as to health services providers that may use such information to determine action items for their day, the data may also be stored in one or more databases that are utilized to help manage health services for patients. For example, the segmentation risk may be used to select patient lists for follow-up with health providers, such as calls or visits.

[0039] Memory **118** may further store interface data **128**. Interface data **128** may include instructions, locations, or other information to retrieve data from or provide data to various systems in communication with the segmentation system **102**, such as care systems **106**. For example, interface data **128** may include types of data stored at a care system **106**, a location of the care system **106**, and specific types of calls or commands that may be used to access the data stored at the care system **106** or other computing system with an interface to the segmentation system **102**.

[0040] The memory **118** may further include (e.g., store or access) instructions for various functions of the segmentation system **102** which, when executed by processor **116**, perform various functions of the segmentation system **102**. The memory **118** may further store data and/or instructions for retrieving data used by the segmentation system **102**. Similar to the processor **116**, memory resources utilized by the segmentation system **102** may be distributed across various physical computing devices. In some examples, memory **118** may access instructions and/or data from other devices or locations, and such instructions and/or data may be read into memory **118** to implement the segmentation system **102**.

[0041] In various embodiments, memory **118** may store instructions for scoring **120**. When executed by the processor **116**, the instructions for scoring **120** generally cause the processor to determine or retrieve (e.g., from patient data **130**) one or more risk scores for a patient or group of patients. Such risk scores may be utilized as input to the categorization model **122** and may be chosen to reduce bias in the categorization model **122**. Risk scores may include, in various examples, hierarchical condition category (HCC) risk scores, chronic illness and disability payment system (CDPS) risk scores, Elixhauser mortality scores, Elixhauser readmission scores, inpatient (IP) and emergency department (ED) likelihood scores, and patient age.

[0042] In some embodiments the IP and ED likelihood scores may come from a database with external data. Alternatively or additionally, the IP and ED likelihood scores may come from a machine learning model, such as one trained on historical data related to IP and ED visits and

readmissions or similar events. For example, the IP and ED likelihood score models may predict six-month likelihoods of inpatient admission and of emergency department visits the patient or group of patients. These models may utilize reconciled data across claims, ADT data, and HIE/EHR sources to predict a comprehensive recent medical history for the patient or group of patients. The models' features, or input variables, include detailed information about recent acute and outpatient utilization, about diagnoses captured and procedures conducted, demographics, and the like. The models may be random forest classifiers that use sample weighting to address class imbalance (e.g., low percentage of positive cases). This helps to ensure that the trained models are not biased to a certain balance of readmissions or other outcomes that may be prevalent due to the skewed training data, e.g., more data may be available for patients who are readmitted as compared to those who are not just due to data collection methods. Similarly, the models may use grid search to optimize hyperparameters. The models may, for example, be trained on 80% of the observations (e.g., index encounters) and performance may be measured on the remaining 20% (e.g., the test set). This again, helps to prevent inadvertent skewing the predictive abilities of the trained models. [0043] The risk scores utilized may reduce bias in the categorization model **122**. The risk scores are generally industry standard risk scores which are validated for healthcare use, are not defined only by past utilization, and are agnostic of healthcare costs. Scores that are defined only by past use or utilization may increase bias as differences in access and trust of the healthcare system impact how different groups interact with the healthcare system. Scores that utilize healthcare costs may increase bias by underestimating need in protected groups. Accordingly, use of industry standard scores which are not defined only by past utilization and which are agnostic of healthcare costs generally reduces bias in the output of the categorization model **122**.

[0044] Memory **118** may further store a categorization model **122**. The categorization model **122** may be an unsupervised clustering algorithm (e.g., k means clustering or other unsupervised algorithms) configured to output clusters of patients based on the patient information provided as input. For example, the categorization model **122** may generate four clusters thought to represent low risk for utilization for physical health, low risk for utilization for behavioral health, high risk for utilization for physical health, and high risk for utilization for behavioral health. In various examples, the number of clusters may vary and may be provided to the categorization model **122** as a changeable parameter.

[0045] The categorization model **122** may utilize various types of patient information as input, generally including at least one risk score for the patient. Such patient information may include, in various examples, patient age, history of hospital admissions, diagnoses, medical history, and the like. Generally, the categorization model also utilizes one or more risk scores generated by scoring **120** and/or stored as patient data **130**. In one example, the categorization model **122** receives, for each patient, a HCC score, CDPS score, Elixhauser mortality score, Elixhauser readmission score, inpatient and ED likelihood scores (either as data received from a database or based on trained ML models). The categorization model **122** may then use a clustering algorithm (e.g., k means clustering) to cluster members into some number of clusters, where each cluster approximates some level of risk for healthcare utilization.

[0046] Instructions for segmentation **124** may further be stored at memory **118**. When executed by the processor **116**, the instructions for segmentation generally utilize output of the categorization model and, in some examples, additional rules or guidelines to assign patients to segments (e.g., risk categories). For example, segmentation may receive clusters or cluster assignments for members from the categorization model **122**. Segmentation **124** may then move individual patients or groups of patients to other clusters, create new groupings, combine clusters into a single grouping, or the like.

[0047] In one particular example, the categorization model **122** may group patients into four groupings including high risk associated with behavioral health, high risk associated with physical health, low risk associated with behavioral health, and low risk associated with physical health.

Segmentation **124** may be configured to group patients into segments representing high risk associated with behavioral health, high risk associated with physical health, rising risk associated with behavioral health, rising risk associated with physical health, and low overall risk.

Segmentation **124** may group patients into segments based on a variety of factors. For example, segmentation **124** may move patients with particular diagnoses from the low risk associated with behavioral health to rising risk associated with behavioral health. Segmentation **124** may move patients from the low risk associated with behavioral health to rising risk associated with behavioral health based on additional factors, such as percentile likelihood of emergency department or inpatient admission. For example, patients above the 75th percentile for likelihood of inpatient admission or patients above the 75th percentile for likelihood of admission to the emergency department may be moved to rising risk associated with behavioral health. Similarly, segmentation **124** may move patients from low risk associated with physical health to rising risk associated with physical health based on patient percentile. For example, patients above the 75th percentile for likelihood of inpatient admission or patients above the 75th percentile for likelihood of admission to the emergency department may be moved to rising risk associated with physical health. Segmentation **124** may further combine low risk associated with behavioral health and low risk associated with physical health into a single low risk category.

[0048] Memory **118** may further store instructions for impact assessment **126**. When executed by the processor **116**, instructions for impact assessment may cause the processor to evaluate patients for particular interventions and prioritize patients for particular interventions. In various examples, impact assessment **126** may utilize an impact model to determine an impact of a particular intervention for a patient based on various patient data **130** associated with the patient. For example, an impact model may be a machine learning model configured to output an impact score or other indication of potential impact based on an input of a diagnosis of a patient. In some examples, impact assessment **126** may be used to assess potential impact of an intervention for a single patient at a time. In some examples, impact assessment **126** may further prioritize multiple patients. For example, impact assessment **126** may obtain impact scores for multiple patients and may rank the multiple patients, with those more likely to be positively impacted by an intervention ranked higher. In some examples, the impact model may utilize unsupervised learning techniques (e.g., clustering) to prioritize patients in a group of patients. In some examples, the impact model may be trained using supervised learning techniques to generate impact scores based on patient data **130**.

[0049] Instructions for user interface configuration **132** may be stored at the memory **118**. When executed by the processor **116**, the instructions for user interface configuration **132** cause the processor **116** to configure various user interfaces to display information generated by the segmentation system **102** and to transmit such user interfaces for display at user interfaces **134** of various user devices **104**. For example, user interface configuration **132** may receive a patient category or segment from segmentation **124** and may configure a user interface displaying the category or segmentation for a patient or group of patients. User interface configuration **132** may similarly receive various information from patient data **130** in order to configure various user interfaces for presentation at user interfaces **134** of user devices **104** in communication with the segmentation system **102**.

[0050] FIG. 3 illustrates an example user interface **200** of the segmentation system **102**. The user interface **200** may generally be fully or partially configured by user interface configuration **132** and may be presented at a user interface **134** of a user device **104** in communication with the segmentation system **102**. The user interface **200** generally shows a user profile of a patient categorized or assigned a segment using the segmentation system **102**. For example, the user interface **200** may display a health summary **202** including an assigned segment or category **204**. The segment **204** shown in the user interface shows that the patient was assigned to a low risk category by the segmentation system **102**. The health summary **202** further shows that the patient

has been placed on a responsive track **206**, which may be a defined track or plan of care for patients in the low risk segment or category.

[0051] In various examples, elements such as the segment **204**, track **206**, and utilization status **208** may be colored or patterned according to a patient's category. For example, in the user interface **200**, the track **206** may be green to illustrate that the track is associated with low risk patients and requires less attention from the user. Similarly, the segment **204** may be colored green or otherwise patterned to indicate that the patient is in the low risk segment. Other segments may be colored or patterned differently. For example, a rising risk segment may be colored orange or yellow, while a high risk segment may be colored red. As shown in the user interface **200**, the patient's utilization status may similarly be colored or patterned based on the patient's level of healthcare utilization.

[0052] The user interface **200** may include additional elements in various examples. Such additional elements may be populated using various patient data **130** stored at the segmentation system **102** and/or accessible by the segmentation system **102** and stored at other data storage **108**. For example, biographical data **210**, appointment data **212**, and treatment pathways **214** are displayed at the user interface **200**. In some examples, the user interface **200** may further include selectable elements to send requests or commands to the segmentation system **102**. For example, via the user interface **200**, a user may request that a segment or categorization for a user be recalculated or updated using the segmentation system **102**. A user may further, through the user interface **200**, update various patient information and such patient information may be provided to the segmentation system **102** to be stored as patient data **130**.

[0053] FIG. 4 illustrates an additional example user interface **300** of the segmentation system **102**. The user interface **300** may generally be fully or partially configured by user interface configuration **132** and may be presented at a user interface **134** of a user device **104** in communication with the segmentation system **102**. For example, the user interface **300** may be utilized by a user responsible for outreach efforts. Such users may utilize information generated by the segmentation system **102** to, for example, develop care plans based on segments generated by the segmentation system **102**, identifying patients who may be good candidates for particular interventions based on impact scores or other measures of impact generated by the segmentation system **102**, and the like. For example, the user interface **300** may display a list of tasks including patients to contact or specific tasks to perform for patients. Such tasks may be based on segmentation or other information generated by the segmentation system **102**. For example, the frequency of contact for a patient may be set based on the assigned segment of the patient. Similarly, the tasks to be performed for each patient may be selected based on either or both of the segmentation or identified interventions generated by the segmentation system **102**.

[0054] The segmentation system **102** may be implemented using various computing systems. Turning to FIG. 5, an example computing system **400** may be used for implementing various embodiments in the examples described herein. For example, processor **116** and memory **118** may be located at one or several computing systems **400**. In various embodiments, user device **104** is also implemented by a computing system **400**. This disclosure contemplates any suitable number of computing systems **400**. For example, the computing system **400** may be a server, a desktop computing system, a mainframe, a mesh of computing systems, a laptop or notebook computing system, a tablet computing system, an embedded computer system, a system-on-chip, a single-board computing system, or a combination of two or more of these. Where appropriate, the computing system **400** may include one or more computing systems; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks.

[0055] Computing system **400** includes a bus **410** (e.g., an address bus and a data bus) or other communication mechanism for communicating information, which interconnects subsystems and devices, such as processor **408**, memory **402** (e.g., RAM), static storage **404** (e.g., ROM), dynamic storage **406** (e.g., magnetic or optical), communications interface **416** (e.g., modem, Ethernet card,

a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network, a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI network), and an input/output (I/O) interface **420** (e.g., keyboard, keypad, mouse, microphone). In particular embodiments, the computing system **400** may include one or more of any such components.

[0056] In particular embodiments, processor **408** includes hardware for executing instructions, such as those making up a computer program. The processor **408** circuitry includes circuitry for performing various processing functions, such as executing specific software for performing specific calculations or tasks. In particular embodiments, I/O interface **420** includes hardware, software, or both providing one or more interfaces for communication between computing system **400** and one or more I/O devices. Computing system **400** may include one or more of these I/O devices, where appropriate. One or more of these devices may enable communication between a person and computing system **400**.

[0057] In particular embodiments, communications interface **416** includes hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computing system **400** and one or more other computer systems or one or more networks. One or more memory buses (which may each include an address bus and a data bus) may couple processor **408** to memory **402**. Bus **410** may include one or more memory buses, as described below. In particular embodiments, one or more memory management units (MMUs) reside between processor **408** and memory **402** and facilitate access to memory **402** requested by processor **408**. In particular embodiments, bus **410** includes hardware, software, or both coupling components of the computing system **400** to each other.

[0058] Accordingly to particular embodiments, computing system **400** performs specific operations by processor **408** executing one or more sequences of one or more instructions contained in memory **402**. For example, instructions for scoring **120**, the categorization model **122**, segmentation **124**, and impact assessment **126** may be contained in memory **402** and may be executed by the processor **408**. Such instructions may be read into memory **402** from another computer readable/usable medium, such as static storage **404** or dynamic storage **406**. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions. Thus, particular embodiments are not limited to any specific combination of hardware circuitry and/or software. In various embodiments, the term “logic” means any combination of software or hardware that is used to implement all or part of particular embodiments disclosed herein.

[0059] The term “computer readable medium” or “computer usable medium” as used herein refers to any medium that participates in providing instructions to processor **208** for execution. Such a medium may take many forms, including but not limited to nonvolatile media and volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as static storage **404** or dynamic storage **406**. Volatile media includes dynamic memory, such as memory **402**.

[0060] Computing system **400** may transmit and receive messages, data, and instructions, including program, e.g., application code, through communications link **418** and communications interface **416**. Received program code may be executed by processor **408** as it is received, and/or stored in static storage **404** or dynamic storage **406**, or other storage for later execution. A database **414** may be used to store data accessible by the computing system **400** by way of data interface **412**. For example, interface data **128** and patient data **130** may be stored using a database **414**. In various examples, communications link **418** may communicate with, for example, user devices to display user interfaces to the segmentation system **102**.

[0061] FIG. **6** illustrates an example process **500** for determining risk categories to patients using the segmentation system **102**. At block **502**, the segmentation system **102** obtains patient information including a risk score. In some examples, the patient information may be obtained from an external source. For example, patient information may be obtained from one or more care

systems **106** and/or one or more data storage locations. Generally, the segmentation system **102** may utilize interface data **128** to obtain such information from various locations. For example, the segmentation system **102** may utilize interface data **128** to make a call to a care system **106** to obtain patient information. Further, some patient information may be stored at the segmentation system **102** as patient data. In some examples, one or more of the risk scores used by the segmentation system **102** may be obtained as patient information (e.g., from a care system **106**). One or more of the risk scores used by the segmentation system **102** may further be calculated by the segmentation system **102** based on other patient information obtained at block **502**. For example, scoring **120** may generate one or more risk scores for a patient or patients based on patient information. In some examples, the risk scores may be stored as patient data **130** for later use by the segmentation system **102**.

[0062] In some examples, the segmentation system **102** may obtain patient information for one patient at block **502**. For example, the segmentation system **102** may receive a request (e.g., through a user interface to the segmentation system **102**) to determine a segment or categorization for a single patient. In other examples, the segmentation system **102** may obtain patient information for multiple patients. For example, the segmentation system **102** may update segments for an entire group of patients at various intervals (e.g., every month) and may obtain updated patient information for each patient in the group of patients when updating the segments.

[0063] The segmentation system **102** clusters patients based on the patient information at block **504**. The segmentation system **102** generally uses the categorization model **122** to cluster patients based on the patient information. For example, patient information for each patient to be included in the clustering may be provided to the categorization model **122** as input. The categorization model **122** may use various types of clustering algorithms to cluster patients into any number of clusters. In one example, the categorization model **122** uses k-means clustering to generate four patient clusters, where the patient clusters generally represent patients having a low behavioral health risk, patients having a low physical health risk, patients having a high behavioral health risk, and patients having a high physical health risk. The number of clusters may generally be provided as input to the categorization model **122** and may be based on the desired number of segments or categories of patients or other factors.

[0064] At block **506**, the segmentation system **102** assigns each patient to a risk category based on the clustering. Segmentation **124** may utilize the clustering from the categorization model **122** to assign patients to a risk category. In some examples, the risk categories may correlate directly with the clusters generated by the categorization model **122**. That is, each cluster may correspond to a risk category or segment. In other examples, segmentation **124** may utilize the clusters as a starting point for segmentation into more or fewer segments or categories than the number of clusters. Segmentation **124** may apply various rules to categorize patients based on the clustering. For example, where the categorization model **122** groups patients into four categories roughly representing high behavioral health risk, high physical health risk, low behavioral health risk, and low physical health risk, segmentation **124** may combine the low behavioral health risk and low physical health risk clusters to obtain one low risk category. Segmentation **124** may further move patients from the low risk category to rising behavioral health risk or rising physical health risk based on various factors including, for example, specific diagnoses. Accordingly, segmentation **124** may generate a different number of segments or categories than the number of clusters generated by the categorization model **122**.

[0065] In various examples, the segments or categories generated using the process **500** may be presented to a user (e.g., at a user device **104** via a user interface **134**). For example, a patient profile may include a patient risk category (e.g., as shown in the user interface **200**). In some examples, the segment or risk category may further be used to configure other aspects of a user interface. For example, the segmentation system **102** may output a recommended action based on an assigned risk category. For example, the user interface may output a recommended check in

with the patient based on risk category, a notification, or other tag that may be used to generate notifications or action items within a health care platform or system.

[0066] In some examples, the segments or categories may be stored as patient data **130** and/or stored at a database **108**. In such examples, users may be tagged with a risk category and the risk category may be displayed after a triggering event, such as hospital admission, a request to view a patient profile, a medical appointment, or the like. The segments may further be utilized by end users to, for example, establish a schedule for contact with the patient, a care plan for the patient, identify particular interventions for the patient, and the like.

[0067] FIG. 7 illustrates an example process **600** for assessing impact of an intervention for a patient using the segmentation system **102**. At block **602**, the segmentation system **102** obtains patient information including a diagnosis. Such patient information may be stored as patient data **130** and/or at a database **108** remote from the segmentation system **102**. In some examples, patient information may further be obtained by querying one or more care systems **106**. For example, a diagnosis may be obtained from a care system **106**, such as an EHR associated with a healthcare provider while other information may be obtained from another care system **106** associated with another healthcare provider, an insurance company, or the like. In such examples, interface data **128** may be used to query various care systems **106** to obtain the relevant patient information.

[0068] The segmentation system **102** determines an impact of an intervention based on the diagnosis at block **604**. In various examples, an intervention may be a program specifically targeted towards improving a particular diagnosis or condition. Accordingly, patients having the diagnosis or condition may be identified as candidates for the intervention or program. In some examples, additional screening and/or analysis may identify a subset of patients having a diagnosis who are more likely to benefit from a particular intervention.

[0069] The impact of the intervention may be based on the diagnosis and various other factors, such as the severity of the diagnosis, control of the condition, comorbidities, historical participation of the patient in various interventions, and the like. For example, a patient with well controlled diabetes may be less likely to experience health improvements from a program aimed at lifestyle interventions for control of diabetes. In another example, patients with significant comorbidities, including terminal diagnoses, may be less likely to experience health improvements from such programs. Further, patients who have already participated in similar interventions may be less likely to experience health improvements from repeating the same or a similar program. Such additional factors may be derived from various patient information, including patient information previously stored as patient data **130**. In various examples, the impact of a diagnosis may be based on a numerical impact score, ranking of the patient, or the like. The impact may include a trained ML model that predicts the likelihood of a positive impact based on the various characteristics of the patient and the diagnosis.

[0070] At block **606**, the segmentation system **102** prioritizes a patient for the intervention based on the determined impact. In various examples, prioritization may include ranking patients from those most likely to benefit from an intervention to those least likely to benefit from an intervention. Such rankings may be displayed, in various examples, via a user interface **134** of a user device **104** accessing the segmentation system **102**. For example, an end user responsible for contacting patients for inclusion in a program or intervention may be provided with a listing of patients to contact and may first contact patients who are more likely to be positively impacted by participation in such an intervention. Accordingly, such programs are more likely to improve health outcomes for patients included in the programs. Further, such prioritization improves resource allocation, as including the most impacted patients is most likely to reduce healthcare spending on management of various medical conditions. In some embodiments, the patient ranking or some grouping of patients (e.g., the highest risk 25% of a patient population) may be transmitted to other databases, such as to other health care providers or the like.

[0071] FIG. 8 illustrates an example process **700** for assigning a risk category to a patient as

assessing an impact of an intervention for the patient using the segmentation system **102**. At block **702**, the segmentation system **102** obtains patient information including a risk score and a diagnosis. Such patient information may be obtained from stored patient data **130** and/or from external sources. For example, patient information may be stored at one or more databased **108** and/or may be obtained from a care system **106**. For example, interface data **128** may be used by the segmentation system **102** to query a care system **106** or multiple care systems **106** to obtain patient information. In some examples, scoring **120** may further generate one or more risk scores at block **702**, and such risk scores may be included in patient information. The risk scores may, in some examples, be stored as patient data **130**.

[0072] The segmentation system **102** clusters patients based on the patient information at block **704**. Generally, the categorization model **122** may cluster patients based on patient information. The segmentation system **102** may cluster the patients using the same or similar methods as those described with respect to block **504** of the method **500**. For example, an unsupervised learning algorithm, such as k-means clustering, may be used to generate a number of patient clusters representing differing levels of patient risk.

[0073] At block **706**, the segmentation system **102** assigns each patient to a risk category based on the clustering. Generally, segmentation **124** utilizes the clustering from the categorization model **122** to assign patients to a risk category. The segmentation system **124** may assign patients to risk categories using the same or similar methods as those described with respect to block **506** of the method **500**. For example, segmentation **124** may combine clusters, create new categories by moving particular patients into new risk categories, and the like.

[0074] The segmentation system **102** identifies an impact of at least one intervention based on the diagnosis at block **708**. In some examples, the segmentation system **102** may evaluate patients in one or several risk categories for a particular intervention without evaluating patients in other risk categories. Such evaluation by risk category may reduce time and processing resources used to evaluate impact of various interventions. For example, when evaluating impact of an intensive program for treatment of behavioral disorders, the segmentation system **102** may first evaluate those patients in the high risk category for behavioral health and may not evaluate patients in the low risk category. In some examples, the segmentation system **102** may first evaluate patients in the high risk category before evaluating patients in lower risk categories.

[0075] At block **710**, the segmentation system **102** prioritizes a patient based on the risk category and the impact. The segmentation system **102** may prioritize patients based on impact using similar or the same methods described with respect to block **606** of the process **600**. The segmentation system **102** may further prioritize based on risk category. For example, higher risk category patients may be prioritized over lower risk category patients, where the prioritization may be within a services lists, database, or the like. In some examples, prioritization may include ranking patients based on impact and/or risk category or segment. Such rankings may be displayed, in various examples, via a user interface **134** of a user device **104** accessing the segmentation system **102**. For example, the user interface **300** may display patients on a “to-do list” based on the prioritization, e.g., a list of patients for immediate follow-up by a service provider (health worker). The rankings may also be stored in a services or health services database utilize to assign and schedule health services.

[0076] In various examples, the segmentation system **102** may interface with other software (e.g., care management software) that uses the risk categories generated by the segmentation system **102**. For example, the segmentation system **102** may provide risk categories to systems utilized to manage patient care. In such examples, such systems may suggest specific interventions or actions based on a patient's risk category. Such systems may further generate alerts based on a patient's risk category or changes in the patient's risk category. For example, an alert may be triggered notifying a patient's primary healthcare provider when the segmentation system **102** categorizes the patient as high risk.

[0077] According to the above examples, the segmentation system **102** may help to ensure that outreach efforts, interventions, care management, and other health programs are carried out in a manner that increases health outcomes for patients while reducing cost of administering such programs. For example, high risk patients may be identified for more intensive care management, improving their health outcomes. Similarly, low risk patients may be identified and provided with less intensive care management, reducing the costs associated with providing healthcare to low risk patients. Currently such patient risk identification and services triaging is not possible given variation between different providers, patient characteristics, and extensive databases.

[0078] The technology described herein may be implemented as logical operations and/or modules in one or more systems. The logical operations may be implemented as a sequence of processor-implemented steps directed by software programs executing in one or more computer systems and as interconnected machine or circuit modules within one or more computer systems, or as a combination of both. Likewise, the descriptions of various component modules may be provided in terms of operations executed or effected by the modules. The resulting implementation is a matter of choice, dependent on the performance requirements of the underlying system implementing the described technology. Accordingly, the logical operations making up the embodiments of the technology described herein are referred to variously as operations, steps, objects, or modules. Further, it should be understood that logical operations may be performed in any order, unless explicitly claimed otherwise or a specific order is inherently necessitated by the claim language.

[0079] In some implementations, articles of manufacture are provided as computer program products that cause the instantiation of operations on a computer system to implement the procedural operations. One implementation of a computer program product provides a non-transitory computer program storage medium readable by a computer system and encoding a computer program. It should further be understood that the described technology may be employed in special purpose devices independent of a personal computer.

[0080] The above specification, examples, and data provide a complete description of the structure and use of exemplary embodiments of the invention as defined in the claims. Although various embodiments of the claimed invention have been described above with a certain degree of particularity, or with reference to one or more individual embodiments, it is appreciated that numerous alterations to the disclosed embodiments without departing from the spirit or scope of the claimed invention may be possible. Other embodiments are therefore contemplated. It is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative only of particular embodiments and not limiting. Changes in detail or structure may be made without departing from the basic elements of the invention as defined in the following claims.

Claims

1. A computer implemented method for providing health services comprising: obtaining from a patient database patient information for a plurality of patients, the patient information including at least a risk score; clustering by a processor the plurality of patients based on the patient information using a categorization model, wherein the categorization model is trained using unsupervised learning; assigning by the processor a risk category to each patient based on the clustering; and outputting to a computing device a recommended action based on the assigned risk category.
2. The method of claim 1, wherein obtaining the patient information comprises accessing the patient information at least one remote health care database and wherein the patient information includes a plurality of risk scores.
3. The method of claim 1, wherein the risk score is based on insurance claims data.
4. The method of claim 1, wherein the risk score is agnostic to healthcare costs.
5. The method of claim 2, further comprising calculating at least one risk score of the plurality of

risk scores based on the patient information.

6. The method of claim 1, further comprising prioritizing the plurality of patients based on the risk category assigned to each patient.

7. A method for managing health services comprising: obtaining patient information for a plurality of patients, the patient information including at least a risk score and a diagnosis for each patient; clustering the plurality of patients based on the patient information using a categorization model, wherein the categorization model is trained using unsupervised learning; assigning a risk category to each patient based on the clustering; identifying, for each patient, an impact of at least one intervention based on a diagnosis and using an impact model; and prioritizing in a health services database at least a patient of the plurality of patients based on the risk category and the impact.

8. The method of claim 7, wherein obtaining the patient information comprises access the patient information at least one remote health care system.

9. The method of claim 7, wherein the risk score is based on insurance claims data.

10. The method of claim 7, wherein the risk score is agnostic to healthcare costs.

11. The method of claim 7, wherein the patient information includes a plurality of risk scores.

12. The method of claim 7, wherein the impact is identified further based on a severity of the diagnosis.

13. One or more non-transitory computer readable media encoded with instructions which, when executed by one or more processors of a segmentation system, cause the segmentation system to: obtain patient information for a plurality of patients, the patient information including at least a risk score; cluster the plurality of patients based on the patient information using a categorization model, wherein the categorization model is trained using unsupervised learning; and assigning a risk category to each patient based on the clustering.

14. The one or more non-transitory computer readable media of claim 13, wherein the instructions further cause the one or more processors to access the patient information at least one remote health care system.

15. The one or more non-transitory computer readable media of claim 13, wherein the risk score is agnostic to healthcare costs.

16. The one or more non-transitory computer readable media of claim 15, wherein the patient information includes a plurality of risk scores.

17. The one or more non-transitory computer readable media of claim 16, wherein the instructions further cause the one or more processors to calculate at least one risk score of the plurality of risk scores based on the patient information.

18. The one or more non-transitory computer readable media of claim 13, wherein the patient information includes a diagnosis, wherein the instructions further cause the one or more processors to identify, for a patient of the plurality of patients, an impact of at least one intervention based on the diagnosis and using an impact model.

19. The one or more non-transitory computer readable media of claim 13, wherein the instructions further cause the one or more processors to prioritize the patient based on the risk category and the impact.

20. A computer implemented method comprising: obtaining patient information for a plurality of patients, the patient information including at least a diagnosis; determining, using an impact model and based on the diagnosis, an impact of at least one intervention based on the diagnosis; and prioritizing in a health services database at least a patient of the plurality of patients based on the impact of the at least one intervention.

21. A computer implemented method comprising: receiving a request for a patient profile; determining a risk category for the patient based at least on patient information associated with the patient, the patient information including at least a risk score, wherein the risk category is determined using an unsupervised machine learning model; and presenting, via a user interface, the patient profile, wherein the patient profile includes the risk category for the patient.

